Topic 6 — Agriculture

Covers syllabus (click to expand)

Crops: Rice | Wheat | Millets | Pulses | Oilseeds | Cotton | Jute | Sugarcane | Tea | Coffee | Rubber | Spices | Fruits | Major GI Crops **Systems & regions:** Cropping Seasons | Patterns | Intensity | Agricultural Regions | Agro-climatic Zones | Irrigation Types | Command Area Development | Seed Production | Agricultural Universities | ICAR | Pest Control | Animal Husbandry | Shifting Cultivation | Organic Farming | Precision Farming | MSP | MSP & Agricultural Marketing | Green Revolution (हरित क्रांति) **Revolutions:** Green | White | Blue | Yellow | Golden | Silver (रूपक) | Pink | Rainbow | Evergreen

Sources baked in: NCERT Class 10 (Agriculture), Class 12 (Resources), cropping patterns + crop conditions + GR/BGREI + irrigation, ICAR/Planning Commission zones, **Ghatnachakra Agriculture CA–195+** + Irrigation/Canals facts, UPPCS Prelims 2018–2025 **Weight:** ★★★★ — crop triples, ACZ 15, revolutions match, CACP/MSP/FRP, GR personalities, UP cane/potato **Last verified:** August 2026 **Current Affairs:** CCEA MSP — **22 mandated crops**; CACP recommends → Cabinet; Kharif 2026–27 common paddy **₹2,441/q**; MSP ≥ **1.5×** cost (Budget 2018–19)

Current Affairs (this topic)

Year	Fact	Why asked	Source
May 2026	CCEA raised MSP for **14 Kharif (खरीफ)** crops, MS 2026–27; common paddy **₹2,441/q** (+₹72); Grade A ₹2,461	Latest paddy MSP figure	PIB / CCEA
Static rule	MSP for **22 mandated crops**; CACP recommends, Cabinet decides; target **≥1.5×** all-India cost (Budget 2018–19)	Institution > old rupees	PIB
2025–26 Kharif	Highest MSP hike then: nigerseed, ragi, cotton, sesamum; extra push pulses/oilseeds/**Shree Anna**	Diversification narrative	PIB
Ongoing	**PM-AASHA** (PSS via NAFED/NCCF); **e-NAM**; **PM-KISAN**; **PMFBY**	Scheme names	DA&FW
2025	Delhi **bio-decomposer** (fungal mix, free) for stubble	Free fungal mix for stubble-to-manure	PYQ / GNCTD

N.1 Land Use, Farming Types, Seasons

India's land-use reporting (NCERT / Ministry of Agriculture) splits reported area into seven heads. These are accounting classes for the agricultural year — not a soil map.

Class	What it means
Forests	Area reported as forest
Barren and unculturable land	Rocky, desert, or otherwise unfit for crops
Permanent pastures and grazing land	Grasslands kept for grazing
Miscellaneous tree crops and groves	Orchards, groves, and tree crops outside net sown
Culturable waste	Land that could be ploughed but is currently unused
Fallow (current + other)	Cultivable land left unsown for a season or longer
Net sown area (NSA)	Land sown with crops **at least once** in the year

- **Net sown area (NSA)** counts a plot **once**, even if it grows two or three crops that year.
- **Gross cropped area (GCA)** counts **every** sowing on that land. One hectare sown with wheat after rice adds **two** to GCA and only **one** to NSA.
- **Cropping intensity** is (GCA ÷ NSA) × 100.
- Example: NSA of 100 ha and GCA of 156 ha give intensity **156%**.
- Intensity rises when irrigation and short-duration varieties allow a second or third crop on the same land.

Farming types

- **Primitive subsistence** farming grows food mainly for the family and includes **jhum** (shifting cultivation) in India's hills and North-East.
- **Intensive subsistence** farming uses small holdings with high labour on rice or wheat for household use and a thin local surplus.
- **Commercial grain** farming aims at market surplus; Green Revolution wheat belts are the Indian face of this type.
- **Plantation** farming is estate monoculture of perennials such as tea, coffee and rubber (**tea** is the usual plantation-crop pick).
- **Mixed farming** keeps crops **and** livestock on the same farm — not merely two field crops together.
- **Dairy** farming is commercial milk production for market sale.
- **Mediterranean** farming (world map) sits under winter rain and grows citrus, olives and vines.
- **Truck farming** grows vegetables for city markets on urban (नगरीय) fringes.
- **Dryland farming** is rainfed on low-rain tracts and favours millets and pulses.
- **Wetland / irrigated farming** uses standing water or assured canal or tube-well supply for rice and sugarcane.
- **Contract farming** links growers to buyers under agreement; **Punjab** is India's classic pioneer state for this model.

Type	Recall
Primitive subsistence	Family food; includes **jhum**

Type	Recall
Intensive subsistence	Small farms, high labour, rice/wheat
Commercial grain	Market surplus (GR wheat belts)
Plantation	Estate perennial — tea/coffee/rubber
Mixed farming	Crops **and** livestock together
Dairy	Milk commercial
Mediterranean (world)	Winter rain; citrus/olives/vines
Truck farming	Vegetables near cities
Dryland farming	Rainfed millets, pulses
Wetland / irrigated	Standing water or canal/tube-well
Contract farming	**Punjab** pioneer in India

- Agriculture still engages about **half or more** of India's workforce (~**54.6%** in Economic Survey **2021–22** / Census 2011) and contributes under **one-fifth** of GVA (~**18.8%** GVA, 2021–22 1st AE). The direction matters more than any single year figure.
- Indian agriculture is marked by over-dependence on nature, low productivity, crop diversity, and **predominance of small holdings** — not large farms.
- Low productivity causes: population (जनसंख्या) pressure on land, small holdings, traditional practices, and disguised unemployment. **Cooperative farming** is a remedy, not a cause.
- **Sikkim** has under **10%** land available for agriculture (forest hill state).
- Uttar Pradesh, Punjab and Haryana are major grain states.
- **Uttar Pradesh** often leads total foodgrain volume.

Cropping patterns

- **Mixed cropping** grows two or more crops **together** on the same field at the same time.
- **Intercropping** mixes crops in fixed **row ratios** so spacing stays planned.
- **Parallel cropping** grows two crops in parallel rows that do **not** compete hard for nutrients; the classic pair is **wheat + mustard**.
- **Double cropping** takes two or more crops in **one crop year** on the same land in sequence — they need not grow side by side.
- **Crop rotation** changes the crop grown on the same land across seasons or years to rest soil and break pests.
- **Sequential cropping** sows one crop after another on the same field in the same year.
- **Relay cropping** sows the second crop **before** the first is harvested so the seasons overlap briefly.
- **Ratoon cropping** raises a new crop from the **stubble** of the previous harvest; **sugarcane** is the classic Indian case.
- **Multiple cropping** means more than one crop in a year on the same land and raises cropping intensity.

- Do **not** swap double cropping (year sequence) with intercropping (same-time proximity).

Pattern	Recall
Mixed cropping	Two+ crops **together** at once
Intercropping	Fixed **row ratios**
Parallel cropping	Parallel rows; classic **wheat + mustard**
Double cropping	Two+ crops in **one crop year** (sequential)
Crop rotation	Sequential change on same land
Sequential cropping	One after another same year
Relay cropping	Second sown **before** first harvest
Ratoon cropping	From **stubble** (sugarcane)
Multiple cropping	More than one crop / year → intensity

- Cropping **pattern** is the share and sequence of crops in a region.
- Climate, soil, irrigation, markets, technology and MSP all shape it.
- Before **1965**, farming was largely rainfed subsistence with coarse cereals and pulses and low yield.
- From **1965–90**, Green Revolution rice–wheat HYV belts spread, then **monoculture** and water stress followed.
- After **1991**, horticulture, commercial crops and contract farming grew.
- Cropping intensity rose from about **111% (1950–51)** to about **156%** now.
- Irrigation and HYV made the extra sowings possible.
- About **86%** of holdings are small or marginal (Agriculture Census **2015–16**: small + marginal ≈ **86.2%** of farmers but only about **47%** of operated area).
- That caps machines and diversification.
- Holding size classes run from **marginal** (under **1 ha**) through small (1–2 ha), semi-medium (2–4 ha), and medium (4–10 ha) to large (**over 10 ha**).
- Average size of operational holdings is often largest in **Rajasthan** (थार) among major states.
- About **half of NSA** is still rainfed.
- Groundwater waters about **two-thirds** (ते-भागा) of the irrigated area.
- Land-use ballpark (Ministry of Agriculture): net sown area about **45–47%**, forest about **23%**, other uses about **30–31%**. A common round figure is **47 / 23 / 30**.
- Horticulture output has overtaken foodgrain tonnage in recent years. That is volume, not calorie king.
- MSP and free power still pull **paddy in Punjab** and **cane in drought (सूखा) Maharashtra**. That is policy, not climate.

Seasons

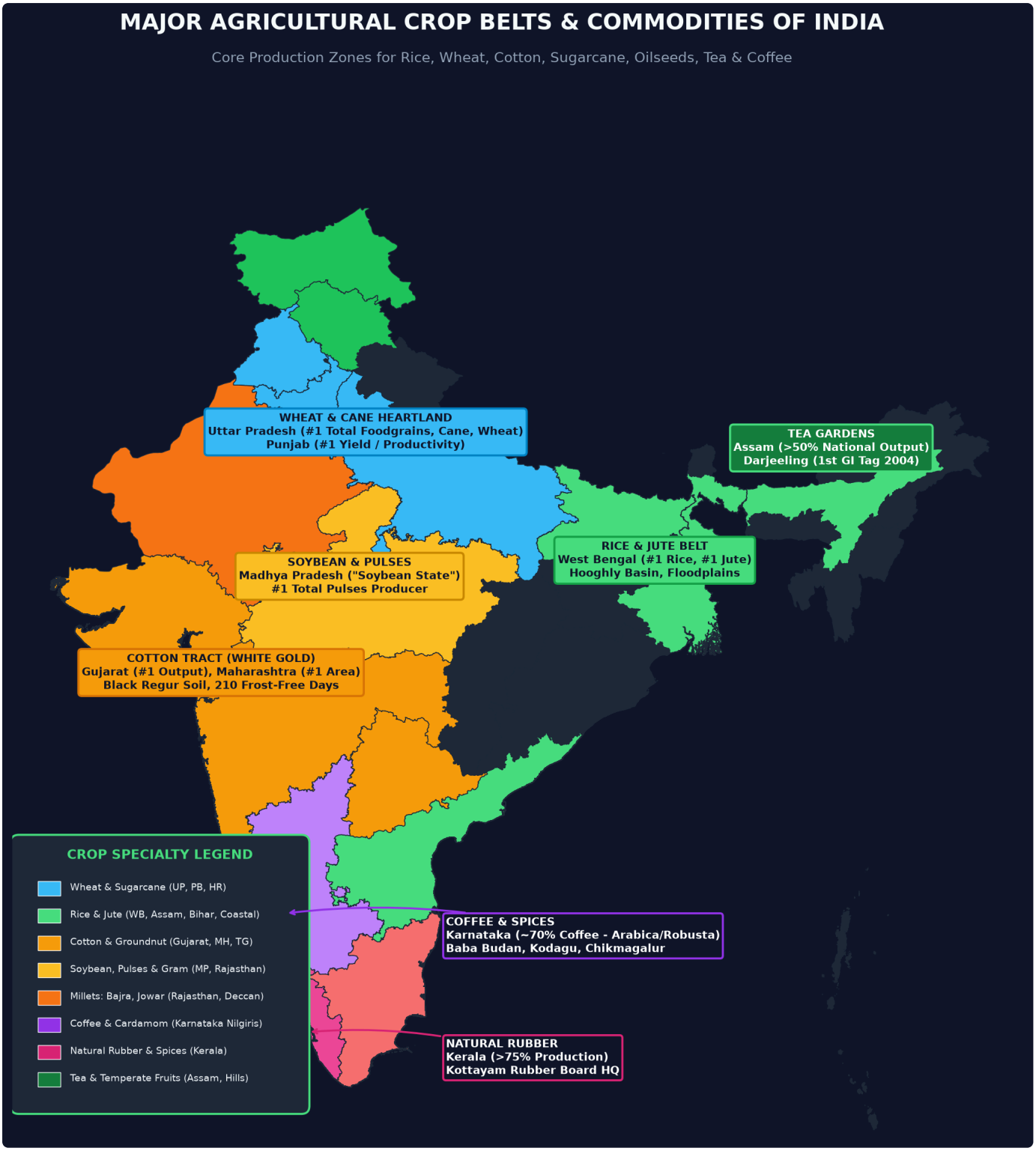
Season	Months	Typical crops
Kharif	Jun–Oct (SW monsoon (दक्षिण-पश्चिम))	Rice, maize, jowar, bajra, ragi, cotton, jute, groundnut, soybean, tur, sesame
Rabi	Oct–Mar	Wheat, barley, gram, mustard, peas, linseed
Zaid	Mar–Jun	Watermelon, cucumber, fodder, vegetables
Long / perennial	—	Sugarcane (10–18 months); tea, coffee, rubber

- India's tropical monsoon supports diversified cropping almost year-round — kharif, rabi and zaid on the same land where water allows.
- India has a **higher share** of its geographical area under cultivation than the **USA, China or Japan**. That is a **percentage-of-land** fact.
- It does **not** mean India has the world's largest **absolute** cropland. The USA's arable area is comparable or larger because the USA's total land is much bigger.
- Rabi belts of PB–HR–W UP also benefit from **western disturbances** (winter rain) plus irrigation.

School agricultural regions (crop-system map — separate from the 15 ACZ names):

- The east is a wet-rice region.
- The plains are a wheat–sugarcane region.
- The Deccan is a cotton region.
- The drylands are a millet region.
- The Ghats are a plantation region.
- The arid west is a livestock region.

N.2 Food Crops



Rice

Rice needs heat plus moisture (or irrigation). India is usually the world's **2nd** rice producer after **China**. In **2024–25**, the Union Agriculture Minister claimed India had become **#1** — treat that as a separate recent claim, not the usual textbook rank.

- Temperature typically runs about **20–27°C** (NCERT often stresses **above 25°C** with high humidity).
- NCERT rain need is about **100 cm or more**. Many notes use **above 150 cm** for the humid core.

- Punjab rice depends on **irrigation**, not on 150 cm of rain.
- Soil is typically **clayey or alluvial** that puddles, with standing water.
- The main season in the north is **kharif**. With irrigation, the south can grow rice **almost year-round**.
- The east also grows three paddies — **Aus**, **Aman** and **Boro** — in Assam, West Bengal and Odisha.
- **Aus** is the summer / pre-monsoon rice.
- **Aman** runs from monsoon into winter.
- **Boro** is irrigated winter–spring rice.
- Methods include **transplantation** (nursery then puddled field), **broadcasting** and **drilling**.
- **DSR** sows seed in the field and saves water.
- **AWD** (alternate wetting and drying) also cuts irrigation.
- DSR needs a dry sowing window. Sudden rain after sowing hurts it.
- Usual volume order is **West Bengal > Uttar Pradesh > Punjab**.
- **Punjab** (then Tamil Nadu / Telangana) leads **yield** because of full irrigation.
- Rice covers the **largest cropped area** among foodgrains.
- India is also the world's largest **rice exporter** in recent years (Basmati to West Asia; non-Basmati to Africa / SE Asia).
- **Azolla** is a floating fern that lives with the cyanobacterium **Anabaena** in its leaf cavities.
- In flooded paddy, Anabaena fixes atmospheric nitrogen.
- When Azolla dies and decays, that nitrogen enters the soil.
- So Azolla is a **biofertiliser** for water-logged rice — not an insecticide.
- Other rice biofertilisers taught with Azolla are **blue-green algae**, Azospirillum, and phosphobacteria.
- **Golden rice** is genetically engineered rice whose grain makes **beta-carotene** (provitamin A). Ordinary rice endosperm does not. Link it to **Vitamin A**, not nitrogen, ozone, or petroleum.
- North-west India's Green Revolution staple sequence is **rice in kharif**, then **wheat in rabi**.
- Only a short gap sits between rice harvest and wheat sowing, so farmers often burn leftover straw to clear the field fast.
- That **stubble burning** is strongest in **Punjab–Haryana**.
- **Krishna–Godavari delta** (Andhra Pradesh) is the classic **"Rice Bowl of India"** tag (distinct from the Chhattisgarh plain nickname).
- **System of Rice Intensification (SRI)** uses alternate wetting and drying. It cuts seed need, methane, and electricity use.
- Hybrid Basmati includes **Pusa RH-10**.
- Transplant seed rate for Basmati is about **15–20 kg/ha**.
- Variety names often asked include **Jaya**, **Padma** (पद्मा), **Hansa**, **Krishna** (कृष्णा), **Ratna**, **Barani (बरनी) Deep** and **Pusa Sugandha**.
- **Mahi (माही) Sugandha** is aromatic rice (not wheat).
- **Jawala** is **not** a rice variety in the usual variety lists.
- **Dee-gee-woo-gen** is the rice dwarfing gene (wheat's is **Norin-10**).

Belt	States
Lower Ganga / delta	West Bengal (multi-rice + jute)
Middle Ganga	E-UP, Bihar
Coastal east	Odisha, AP/TG, TN
Irrigated NW	Punjab, Haryana, W-UP
Krishna–Godavari delta	AP — “Rice Bowl of India”
Chhattisgarh plain	“Rice bowl” tag

Wheat

Rabi cereal. India ranks **2nd** globally.

- It is cool while growing (about **10–15°C**).
- It needs a **brighter / warmer** ripening phase (about **20–25°C**).
- It needs **moderate (नरम दल) temperature and moderate rainfall** (not high heat plus heavy rain).
- Frost at flowering or rain at harvest hurts yield.
- Rain of about **50–75 cm** evenly (many notes say about **75 cm**), or irrigation, suits wheat.
- Well-drained **alluvial loam** is the classic soil.
- Two textbook belts stand out: the **Ganga–Satluj** plains of the north-west, and **black-soil** Deccan wheat.
- Core states are **Uttar Pradesh, Punjab and Haryana**, with Madhya Pradesh, Rajasthan and Bihar also strong.
- Pairing **Assam with wheat** is wrong.
- HYV semi-dwarf wheat plus water and fertiliser created the Green Revolution surplus in Punjab–Haryana–western UP.
- The dwarfing gene in wheat is **Norin-10**.
- **Macaroni wheat** (*Triticum durum*) suits **rainfed / dry** conditions.
- **Triticale** is a hybrid of **wheat and rye**.
- Wheat diseases include **yellow, brown and black rust**.
- **Karnal bunt** is fungal (*Tilletia indica*, first noted **1931**).
- Varieties often asked include **Sonalika**, **Arjun**, **Kalyan Sona**, **Sonora-64** (induced mutation at IARI), **Raj 3077**, **Pusa Sindhu Ganga (HD 2967)** and **UP-308**.
- The most critical irrigation stage for wheat is often taught as **CRI (crown root initiation)**, not flowering.
- **UPAS-120** is a pigeon-pea (arhar) variety suited for **double cropping with wheat**.

Maize & millets (nutri-cereals)

- Maize is mostly a **kharif** crop used for food, feed and starch.
- Rabi maize also appears in Bihar pockets.
- The botanical name is **Zea mays**. It is called the **“Queen of cereals.”**

- It is also used for starch, biodiesel (बायो-डीजल) feedstock and alcoholic beverages.
- **Shaktiman-I** and **Shaktiman-II** are genetically modified / hybrid maize varieties.
- Maturity is often about **90–150 days** (options near **110 days** are closest).
- Maize is a **C4** plant (rice and wheat are C3).
- Leading producers in recent data are often **Karnataka, Madhya Pradesh and Maharashtra**.
- Millets are rainfed, drought-tolerant, light-soil crops and are often mixed with pulses.
- The UN marked **2023** as the **International Year of Millets**.
- The Government of India brand is **Shree Anna**.
- The millet group includes sorghum, kodo, kangani and ragi — not moong.

Crop	Leading state	Note
Jowar (ज्वार)	**Maharashtra**	Some rabi jowar in the peninsula
Bajra (बाजरा)	**Rajasthan**	Lowest-rainfall cereal
Ragi	**Karnataka**	Ca-rich; red/lateritic soils OK

Pulses

- **Rhizobium** fixes nitrogen in root nodules of **pulses** and other legumes. It is not the biofertiliser story for rice, wheat, or cane.
- Protein crop that also rebuilds soil fertility.
- Green Revolution historically neglected pulses vs wheat–rice.
- India is the world's **largest** producer — and still a major importer (largest consumer and importer among major crops).
- **Cobalt** is essential for symbiotic nitrogen fixation by Rhizobium (and for vitamin B12 synthesis).
- Balanced NPK ratios for legumes are often **0:1:1**, **1:2:2** or **1:2:3** — not a cereal-heavy N dose.
- About **90%** of pulse area is classically rainfed.
- Arhar varieties often asked include **Malaviya (मालवीय) Chamatkar**, **Bahar** and **Amar**.
- **Aparna** is a leafless pea.
- **UPAS-120** is arhar bred for wheat double-cropping.

Pulse	Season	Note
Gram / chana	**Rabi**	MP often among leaders
Masoor (lentil)	Rabi	Cool season
Tur / arhar	**Kharif**	Central India; red gram origin often tagged **India**
Moong, urad, moth	Kharif	Short-duration; drylands; urad can be kharif and rabi in the south
Peas	Rabi	With wheat belts

States: MP, RJ, MH, UP, KA. Mixed cropping with millets is common.

N.3 Cash, Fibre & Plantation

Oilseeds — Yellow Revolution

Crop	Season	Core state
Groundnut	Kharif	**Gujarat**
Mustard / rapeseed	**Rabi**	**Rajasthan**
Soybean	Kharif	**Madhya Pradesh** (oilseed, not pulse)
Sesame	Kharif	Drylands
Sunflower	Flexible	KA/MH/AP
Castor	Drought-hardy	GJ/RJ
Linseed	Rabi	Central/north
Safflower	Cool dry rabi	Deccan
Niger	Minor	Tribal (आदिवासी)/peninsula

- Oilcake feeds dairy (White Revolution link).
- Do **not** swap **Yellow** (oilseeds) with **Golden** (horticulture and honey).
- Groundnut is the classic **dryland** oilseed.
- **Pegging** after flowering pushes the peg into soil — a groundnut-specific fact.
- Groundnut needs **gypsum**-rich soil for quality pods.
- Sesame often tops oil-content % among common oilseed options.
- **Safflower** is *Carthamus tinctorius*.
- Mustard varieties include **Varuna** (वरुणा), **Pusa Bold** and **Pitambari** (yellow mustard).
- **Kaushal** is a groundnut variety.
- Soybean area is largest in **Madhya Pradesh**.
- Recent production leadership can flip with **Maharashtra** — check the year of the data.

Cotton (white gold)

Kharif fibre. India is among the world's top / often **largest** producers; **Bt cotton** dominates area. It takes about **6–8 months**.

- It needs warmth above about **21°C**, rain about **50–100 cm**, about **210 frost-free days**, and bright sun at boll opening.
- Hard frost kills cotton.
- Best soils are **black regur (रेगुर)**.
- The north zone uses **deep alluvium**.
- The south zone uses mixed black–red soils.
- Three belts stand out: **North** (Punjab–Haryana–Rajasthan–western UP, irrigated), **Central** (Gujarat–Maharashtra–Madhya Pradesh, the volume core), and **South** (Telangana–Andhra Pradesh–

Karnataka–Tamil Nadu).

- About **two-thirds** of Indian cotton is **rainfed**.
- Yield is below the world average.
- Cotton can take some **salinity**.
- It **hates waterlogging**.

Four species India grows (India is often taught as the only country that cultivates all four recognised cotton species):

Species	Common tag	What it gives
Gossypium arboreum	Desi / Old World	Short staple; drought- and pest-hardy
Gossypium herbaceum	Desi / Old World	Short–medium; harsh rainfed tracts (GJ / KA pockets)
Gossypium hirsutum	American / Upland	**Bulk of India's crop**; medium to long staple; most **Bt** hybrids
Gossypium barbadense	Egyptian / Sea Island	**Extra-long staple (ELS)**; small premium pockets in the south

Staple-length classes (fibre length decides mill use; CCI / trade (पण्याध्यक्ष) classes):

Class	Approx. length	India note
Short staple	≤ about **20 mm**	Desi types; coarse cloth, blending, surgical cotton
Medium staple	about **20.5–24.5 mm**	Large share of domestic mill use
Medium-long staple	about **25–27 mm**	Modern mill demand
Long staple	about **27.5–32 mm**	High-speed spinning; MSP is also quoted for a long-staple grade
Extra-long staple (ELS)	about **≥32.5 mm**	Premium yarn; India still **imports** much ELS (Egypt (मिस्र) / USA style)

- India's harvested cotton is still **mostly medium** (and medium-long), not Egyptian-style ELS. Mills **blend** staples.
- MSP for kapas is announced separately for **medium staple** and **long staple** grades (CACP recommends; the Union Cabinet decides).
- **Ahmedabad** (अहमदाबाद) lies in a major cotton **region** (raw (रॉ) material true).
- Traditionally India's largest cotton-textile **centre** is **Mumbai**, not Ahmedabad.
- Cotton is indigenous to India (Rigveda (ऋग्वेद) / Manusmriti (मनुस्मृति) mention). India was first to commercialise **hybrid cotton**.
- Cotton fibre is obtained from the **seed**.
- **Khandwa–Khargone** (Madhya Pradesh) is tagged the **"White Gold"** cotton belt. Maharashtra also calls cotton white gold on black soil.
- Expansion of irrigation made cane more profitable than cotton on parts of the Deccan black-soil belt.

Jute (golden fibre) & mesta

- Hot humid conditions run about **24–35°C**, with rain often cited about **120–150 cm or more**, high humidity, and deltaic **alluvium**.
- Jute needs **retting** in still or standing water.
- It is the second fibre after cotton.
- It is largely **rainfed** and uses little fertiliser or pesticide — the opposite of cotton.
- It is sown to catch the monsoon.
- Many notes give a long **February–October** window (about 8–10 months).
- Still treat it as a **humid-east kharif** crop, not a rabi cereal.
- The core belt is **West Bengal**, with Assam, Bihar and Odisha also important.
- Mills sit on the **Hugli**.
- Pairing **Uttar Pradesh with jute** is wrong.
- India is the world's top jute producer.

Jute types India grows (two true jute species + one related fibre often asked with them):

Type	Botanical name	Field note
White jute	*Corchorus capsularis*	Stronger flood tolerance; classical lowland / waterlogged delta crop
Tossa jute	*Corchorus olitorius*	Softer, usually better fibre and higher yield; prefers **uplands** and hates prolonged flood
Mesta	*Hibiscus cannabinus* / *H. sabdariffa* (related fibre, not true jute)	Grown on **drier** tracts as a jute substitute; coarser fibre

- In volume, eastern India grows both white and tossa; **tossa** has expanded where drainage is better.
- Do **not** call mesta a separate "jute species" in a strict botanical question — it is the **allied fibre** used like jute.

Sugarcane

Long-duration crop (**~10–18 months**) — not a single short season.

- Ideal temperature is about **21–27°C**.
- Water need is about **75–150 cm**, or irrigation.
- Any soil that **holds moisture** works.
- The crop **exhausts** fertility, so manure matters.
- Too much rain **lowers sugar**.
- Too little rain makes a **fibrous** cane.
- A bright, open second half thickens juice.
- A short **cool dry** spell at harvest is ideal.
- **Frost** and the **loo** hurt north Indian cane.
- South India lacks both, so yield is often higher.
- **Ratoon** (stubble sprout) is common.

- It saves planting cost, but yields fall after a cycle or two.
- **Uttar Pradesh** often leads **area and production**.
- **Maharashtra** often leads **yield** and cooperative factories.
- Cane on a small share of Maharashtra land still drinks a huge share of irrigation water.
- Cooperative mills in Maharashtra are a true institutional fact, but they do **not** explain why Uttar Pradesh's yield is lower. North Indian frost, the loo, a shorter crushing season and ratoon cycles do.
- Price uses **FRP** (plus possible state SAP) — not cereal MSP.
- By-products include bagasse and molasses / ethanol.
- Sugar can also come from **beet**, not cane alone.
- Philippines cane and coconut credit go to the **Spanish and Americans**.
- **Uttar Pradesh** is the classic **"Sugar Bowl"** of India by quantity.
- South India often leads **sucrose and productivity** (longer crushing season, frost-free).
- The first sugar mill in India was **Pratappur** (Deoria, UP) in **1903**.
- Sugarcane breeding HQ is the **Sugarcane Breeding Institute, Coimbatore** (1912; under ICAR).
- A key variety is **Co. 1148**.
- **Adsali** cane in the Maharashtra low-rain belt takes about **16–18 months**.
- **Sustainable Sugarcane Initiative (SSI)** (WWF–ICRISAT, **2009**) uses fewer seeds, nursery seedlings, wider spacing, drip-friendly layout and more intercropping scope — it does **not** ban all chemical fertiliser.
- Among pearl millet, red gram, sunflower and sugarcane, **sugarcane** is the **least water-efficient**.

Tea · Coffee · Rubber · Spices

Crop	Climate	Site	India
Tea	20–30°C; 150–300 cm	Well-drained **slopes** (no stagnant water)	Assam (volume), WB **Darjeeling GI**, Nilgiris/TN–KL; **plantation**
Coffee	15–28°C; 150–250 cm	**Shade**, well-drained	**KA > KL > TN** (Baba Budan / Coorg / Wayanad / Nilgiri)
Rubber	25–35°C; **>200 cm**	Humid equatorial; *Hevea brasiliensis* latex	**Kerala** (also TN, KA, A&N, Tripura pockets)
Pepper / cardamom	Hot wet Ghats	Hills	**Kerala** largest
Chilli	Warm Deccan	—	AP/TG strong — **not Kerala-only**
Turmeric / ginger	Warm	South + NE	Many GIs

- **CTC** (Crush–Tear–Curl) is bulk everyday tea.
- **Orthodox** processing makes premium leaf tea.
- Tea likes acidic, well-drained hill soils.
- Tea is often nicknamed **Green Gold**.

- Origin notes tag the **Yunnan** plateau of South China.
- Coffee was first grown in India in **Chikkamagaluru** (Karnataka).
- Coffee is propagated by **seeds**.
- Tea is propagated mainly by **stem cuttings**.
- Kerala is the classic **"Garden of Spices."**
- Black pepper is nicknamed **black gold** or **black diamond**.
- **Clove** is the flower bud of *Eugenia caryophyllata*.
- The **Coffee Board** headquarters is in **Bengaluru**.
- The **Tea Board** headquarters is in **Kolkata**.
- The **Rubber Board** headquarters is in **Kottayam**.
- The **Tobacco Board** headquarters is in **Guntur**.
- The **Spices Board** headquarters is in **Kochi**.
- The **National Horticulture Board** was set up in **1984** with headquarters at **Gurugram**.

Coffee species	Note
Arabica	Higher elevation, milder cup, more pest-sensitive
Robusta	Lower elevation, stronger cup, hardier

- World coffee order often taught for **2016** is **Brazil > Vietnam > Colombia > Indonesia**.
- Cocoa majors are Ivory Coast, Ghana (घन) and Cameroon. Latvia is **not** a cocoa major.
- Philippines cane and coconut history is linked to Spanish and American periods.
- Tobacco leadership has shifted toward **Gujarat** in recent data (older notes often said Andhra Pradesh).
- Coconut leadership still sits with **Kerala** (Karnataka is close in recent years).
- Cashew leadership often sits with **Maharashtra** among states.
- Commercial saffron production is centred in **Jammu & Kashmir** (Zafran).
- **Guar (cluster bean)** gum is used in **shale-gas / hydraulic fracturing**; India–Pakistan dominate world production.
- In sericulture, India is usually world **#2** after China.
- **Mulberry** silk leadership is **Karnataka**.
- **Muga** and **eri** silk leadership is **Assam**.
- **Tropical tasar** leadership is **Jharkhand**.
- **Oak tasar** leadership is **Manipur**.

N.4 Fruits, Potato, GI

- Tropical fruits include mango, banana and pineapple.
- Subtropical fruits include citrus and grapes.
- Temperate hill fruits include apple in Jammu & Kashmir, Himachal (हिमाचल) Pradesh and Uttarakhand (उत्तराखंड).

- The world citrus belt is the **Mediterranean**.
- India's orange renown centres on **Nagpur**.
- Improved **guava** varieties include **Lalit** and **Banarsi**.
- **Sultana, Gulabi and Kali Champa** are **guava** varieties.
- Do not treat this trio as grapes. The name **Sultana** is also used for a seedless grape, and that is the usual mix-up.
- **Uttar Pradesh** is the **leading potato producer**.
- The CIP South Asia Regional Centre is at **Singna, Agra**, not Aligarh.
- Best processing potato varieties are **Kufri Chipsona-2** and Chipsona-3 in the plains, and **Kufri Himsona** in the hills.
- **Sindhu** is a seedless mango.
- **Amrapali** is a Dasheri × Neelam hybrid from IARI (**1971**).
- **Kanchan**, **Krishna** and **Banarasi** (बनारसी) are improved amla (Indian gooseberry) varieties.
- **Lalit** is a guava variety.
- The ginger storage organ is a **rhizome**.
- The **Golden Revolution** covers horticulture **and honey**.
- **Arunachal Pradesh** is the classic low-cost orchid / export-horticulture climate pair.

GI / variety	Place
Darjeeling Tea (दार्जिलिंग चाय)	WB hills
Basmati	North India GI belt
Alphonso	Maharashtra
Banganapalle	AP
Gir (गीर) Kesar	Gujarat
Dasheri / Langra	UP
Malabar Pepper	Kerala
Kashmiri saffron	J&K
Nashik grapes	MH
Coorg coffee / Alleppey cardamom	KA / KL GI note

N.5 Agricultural Regions & Agro-climatic Zones

Three **number** facts — do not mix:

System	Count	Use
Planning Commission **ACZ**	**15** (~72–73 sub-zones)	Broad planning

System	Count	Use
ICAR **NARP**	**~127**	Research / extension
NBSS&LUP **AER**	**~20**	Soils + LGP (Length of Growing Period)

15 Agro-climatic Zones (Planning Commission)

#	Zone	Core	Agri tag
1	Western Himalayan	J&K, HP, UK	Apple, valley cereals
2	Eastern Himalayan	Assam, NE, Sikkim, N WB	Tea, rice, spices, jhum
3	Lower Gangetic Plains	West Bengal	Rice, jute, mustard
4	Middle Gangetic Plains	E-UP, Bihar	Rice–wheat, maize, cane
5	Upper Gangetic Plains	W/central UP	Wheat–rice–cane (GR)
6	Trans-Gangetic Plains	PB, HR, Delhi, parts RJ	Intensive wheat–rice
7	Eastern Plateau & Hills	CG, JH, OD uplands	Rice, millets, rainfed
8	Central Plateau & Hills	MP, RJ, Bundelkhand edge	Soybean, wheat, gram
9	Western Plateau & Hills	MH upland	Cotton, jowar, pulses
10	Southern Plateau & Hills	Interior AP/TG, KA, TN	Groundnut, millets, cotton
11	East Coast Plains & Hills	Coastal OD–AP–TN	Rice, coconut, cashew
12	West Coast Plains & Ghats	KL, coastal KA, Goa (गोवा), Konkan	Rice, coconut, spices, rubber, coffee
13	Gujarat Plains & Hills	Gujarat	Cotton, groundnut
14	Western Dry Region	W Rajasthan	Bajra, pulses, livestock
15	Islands Region	A&N, Lakshadweep (लक्षद्वीप)	Coconut, horticulture

- One state can sit in **multiple** zones.
- Western Dry ≠ West Coast Ghats.
- Do not swap the counts: Planning Commission agro-climatic zones are **15**; NBSS agro-ecological regions are about **20**. A claim that India has **20** agro-climatic and **15** agro-ecological regions reverses both numbers, so it is false.
- **P. Sengupta and G. Sdasyuk (1968)** divided India into **60** micro agricultural regions (Registrar General monograph).

Agro-ecological regions (NBSS&LUP ≈ 20)

Grouped by ecosystem (पारिस्थितिकी तंत्र) (remember the **count** and the **15 vs 20** swap trap more than every name):

Ecosystem	Example AER tags
Arid	Western Himalayas cold arid; Western Plain–Kachchh; Deccan Plateau (दक्कन पठार) arid
Semi-arid	Upper / Middle Ganga plains; Rajasthan–Gujarat plains; Deccan red-black / red sandy
Sub-humid	Eastern Plateau (Satpura (सतपुड़ा)–Mahanadi (महानदी), Bundelkhand, red-laterite (लेटराइट)); Lower Ganga; warm W Himalaya (हिमालय); Bengal Basin
Humid–perhumid	Assam–N Bengal plains; Eastern Himalayas; Purvanchal (पूर्वांचल) hills (पूर्वांचल पहाड़ियाँ)
Coastal	Eastern Coastal Plains + A&N; Western Ghats (पश्चिमी घाट) coastal plains and hills

PYQ — UPPCS Prelims 2022, Q142

Which is correctly matched?

A. Gujarat—Tea

B. Uttar Pradesh—Jute

C. Kerala—Rubber

D. Assam—Wheat

Show answer

Ans: C — Rubber–Kerala is the only correct pair in that set.

N.6 Irrigation & Command Area Development

CAD develops the **command** of an existing project (channels, drainage, warabandi). It is **not** "build a new dam." Classic CADP launches (from **Dec 1974**) include **Sharda (शारदा) Tributary**, **Ramganga** (रामगंगा), and **Gandak** (गंडक).

- Most of India is warm enough that evaporation creates a soil-moisture deficit for crops — hence irrigation.
- Agriculture uses most of India's freshwater (NCERT surface about **89%** / groundwater about **92%**; often cited about **80%** overall).
- In **2018–19**, tubewells covered about **48.5%** of irrigated area.
- Wells plus tubewells together covered about **64%**.
- Canals covered about **23%**.
- Tanks covered about **2.3%**.
- About **two-thirds** of irrigated land drinks **groundwater**.
- Punjab paddy plus free power is the over-exploited poster case.
- Project size by **CCA**: **minor** projects are **2,000 ha or less** and hold about **62%** of irrigation potential (wells, tubewells, tanks, lift, drip, sprinkler).
- **Medium** projects cover **2,000–10,000 ha**.

- **Major** projects exceed **10,000 ha**.
- Major plus medium together cover about **38%**.
- Rice and sugarcane together take a huge share of irrigation water, often in **water-stressed** states.
- Micro-irrigation (drip/sprinkler) is still a **small** slice of irrigated area.
- It cuts nutrient loss and can slow groundwater decline in places — it is **not** the only dryland irrigation method.
- PMKSY (**1 Jul 2015**) carries the “more crop per drop” scheme tag.
- **Protective / life-saving irrigation** means watering at **permanent wilting point (PWP)**.
- The peninsula classic is **tanks and ponds** on hard rock with seasonal rivers.
- Sir **Arthur Cotton** is the pioneer of irrigation works in South India.
- **Uttar Pradesh** leads absolute (निरपेक्ष) tubewell / well irrigated area and replenishable groundwater among major states (about **40.7 bcm** for irrigation).
- UP net irrigated share (**2018–19**): tubewell about **74.6%**, canal about **15.2%**, other wells about **8.8%**, tanks about **0.6%**.
- Spectacular tubewell growth is noted on the **Saryupar** plain, where canals are scarce.

Type	Best terrain	Risk / note
Perennial canal	Alluvial plains (PB–HR–UP; IGC RJ). Clayey beds **leak less**	Waterlogging / usar
Inundation canal	Flood plains	Unreliable timing
River lift	Where canal flow is irregular; common in **peninsular** belts; **Pattiseema** = Godavari→Krishna	Pumping cost
Tube well	Alluvial aquifers; GR NW + W-UP	Groundwater depletion
Dug well	Local / hard-rock	Limited yield
Tank	**Peninsular** hard rock	Siltation
Drip	Horti, cane, cotton, veg; 40–60% water save; less weed / erosion	High efficiency; capital cost
Sprinkler	Sandy / undulating	Wind drift

CADWM pack: field channels, land levelling, **warabandi** (roster turns), drainage, farmer organisations, suitable cropping patterns.

- Over-irrigation without drainage produces saline (लवणीय) or alkaline usar.
- The Indira Gandhi (गांधी) Canal (इंदिरा गांधी नहर) is the desert example.
- Zero / reduced tillage, gypsum before irrigation, and leaving crop residue on the field all help **water conservation** in agriculture.
- The full canal-project map (Gang, IGC, Upper/Lower Ganga, Sharda, Farakka, Hariyali, IWMP) is covered under Lakes and Water Resources.

N.7 Seeds, ICAR, Universities, Pests & Practices

- **ICAR** (HQ **New Delhi**) is the apex research and agri-education body.
- **IARI Pusa** is the flagship institute.
- **CACP** sets price advice — never swap with ICAR.
- Seed purity runs downhill from **Breeder** to **Foundation** to **Certified**.
- Tag colours are Breeder **golden yellow**, Foundation **white**, and Certified **blue**.
- A **KVK** is the district farm-science centre.
- India's **first agricultural university** is **G.B. Pant University of Agriculture and Technology, Pantnagar** (opened as Uttar Pradesh Agricultural University; inaugurated by **Jawaharlal Nehru** (जवाहरलाल नेहरू) on **17 November 1960**).
- Other Green Revolution–era SAUs include **PAU Ludhiana** and **HAU Hisar**.
- UP SAUs include CSAUAT Kanpur (कानपुर) and SVPUAT Meerut (मेरठ).
- **'History of Indian Agriculture'** is by **M.S. Randhawa** (Mohinder Singh Randhawa) — not Swaminathan.
- FAO rule: for safe grain storage, moisture should not exceed about **14%**.
- **Agmark** is quality certification under the **Agricultural Produce (Grading and Marking) Act, 1937**.
- The **Seed Village Concept** trains farmer groups to produce quality seed for self-use and neighbours at the right time and affordable cost (not "ban buying seed" and not "only certified-seed villages").
- Seed Replacement Rate is constrained by a demand–supply gap for quality seed in **low-value, high-volume** crops.
- India **does** have a National Seeds Policy (**2002**).
- Private firms are active in high-value vegetable and horti seed.
- The **Zero Till Seed-cum-Fertilizer Drill** was developed at **G.B. Pant University, Pantnagar**.
- The **Borlaug Award** honours agricultural science (named after Norman Borlaug; from **1972**).

Green manure, nutrients, fertigation

- Green manure means growing plants such as **sunhemp / sanai, dhaincha, guar and lobia (cowpea)** and ploughing them in before the main crop.
- Nitrogen % often taught: cowpea about **0.49%** (highest among common options), sunhemp **0.43%**, dhaincha **0.42%**, guar **0.34%**.
- Sunhemp often returns the **maximum kg N/ha**.
- **Lobia** is preferred green manure on newly improved **arid** land.
- Balanced fertilisers raise production, improve grain quality, and maintain soil productivity.
- Fertiliser **intensity (kg/ha)** leaders are **Punjab and Haryana** (Puducherry often highest among UTs).
- Fertiliser **total tonnage** leader is **Uttar Pradesh**.
- **Fertigation** dissolves fertiliser in irrigation water (usually drip).
- It can raise nutrient availability and cut leaching.
- **Rock phosphate / superphosphate** is a poor fertigation choice because of precipitation.
- Controlling irrigation-water alkalinity is a taught advantage.

- **Permaculture** discourages monoculture, stresses mulching, and resists salinity buildup better than chemical monoculture.
- It **is** workable in semi-arid settings. Do not treat "not easily possible in semi-arid tracts" as true.
- **Zero tillage** allows wheat sowing without burning previous residue, supports direct-seeded rice logic, and aids carbon sequestration (कार्बन पृथक्करण).
- **Conservation agriculture** core practices are minimum / zero tillage, residue mulch on the soil, and crop rotation / sequencing — **not** "ban plantation crops."
- Eco-friendly practice sets often include crop diversification, legume intensification, **tensiometer** use, and **vertical farming**.
- **Green agriculture** (UPPCS wording) means integrated pest management plus integrated nutrient supply plus integrated natural resource management.

Schemes (practice)

Scheme	Key point
Soil Health Card	Soil fertility / balanced nutrient advice
PMKSY (incl. more crop per drop)	Irrigation access / water efficiency
PKVY (Paramparagat Krishi Vikas Yojana)	Organic / traditional farming clusters
Kisan Credit Card	Crop credit via commercial, cooperative and RRBs
e-NAM / national agri market push	Better price discovery
NFSM	Rice, wheat, pulses (later oilseeds / millets) productivity

- **IPM** combines cultural, biological and chemical tools — not calendar spray alone.
- Bt cotton targets bollworm.
- Azolla is a biofertiliser, not an insecticide.
- The 2025 Delhi **bio-decomposer** was free to farmers.
- The fungal mix turns stubble to manure.
- **Butachlor** is a herbicide (versus chlorpyrifos / quinalphos as insecticides and carbendazim as a fungicide).
- The first widely used herbicide is often tagged **2,4-D**.

Disease / problem	Cause
Citrus canker	**Bacteria**
Red rot of sugarcane	**Fungus**
Potato "Krishnakant"	**O_2 deficiency**
Wheat "Sahu"	**Insect**
Wheat Karnal bunt	Fungus (*Tilletia indica*)
Wheat rusts	Yellow / brown / black

Disease / problem	Cause
Paddy Khaira	Zn deficiency ($ZnSO_4$ spray)
Potato late blight	Classic potato disease
Arhar wilt	Wilt disease

N.8 Animal Husbandry

- India is the world's largest **milk** producer by volume (about one-fourth of world milk in recent FAOSTAT figures).
- Buffalo are important in the north (higher fat).
- The **White Revolution** (श्वेत क्रांति) is linked to **Verghese Kurien**, **Operation Flood** (ऑपरेशन फ्लड), **NDDB**, and **Amul / Anand** cooperatives.
- Textbook phases run **I 1970–81** (milk grids / coops), then **II 1981–85** (expand dairies), then **III 1985–96** (consolidate).
- White Revolution launch is often tagged **July 1970**.
- The **Indian Dairy Corporation** was set up at **Anand (Gujarat)** in **1970**.
- **NDDB** was established in **1965**.
- **NDRI** is at **Karnal** (Deemed University from 1989).
- The **Silver Revolution** covers eggs and poultry.
- The **Blue Revolution** (नीली क्रांति) covers fisheries and aquaculture (**Hiralal Chaudhuri** often tagged).
- Recent inland plus marine fish leadership often sits with **Andhra Pradesh**.
- Milch cows give more milk but weaker draft (Gir, Sahiwal, Red Sindhi, Tharparkar).
- Drought breeds work better as draft (Nagauri, Hallikar, Kangayam, Khillari).
- **Tharparkar** is a Rajasthan border-region cattle breed.
- **Hallikar** is Karnataka — not a Rajasthan breed.
- Goat is the "poor man's cow."
- **Jamunapari** is India's classic high-milk goat.

Breed	Animal	Tag
Murrah	Buffalo	Haryana / high yield
Bhadawari	Buffalo	UP / Bundelkhand note
Gir	Cattle	Gujarat
Sahiwal	Cattle	Punjab-region note; high milk
Tharparkar	Cattle	Western Rajasthan border
Deoni	Cattle	Dual-purpose
Jamunapari	Goat	UP renown; high milk

Breed	Animal	Tag
Barbari / Beetal	Goat	Other common breeds

- 20th Livestock Census (**2019**): cattle about **193 million**. Top cattle states often **West Bengal, UP, Madhya Pradesh**. Top total livestock often **UP > Rajasthan > MP**. Top sheep often **Telangana**.
- UP usually leads state milk production; Rajasthan and Madhya Pradesh follow in recent years.

N.9 Jhum, Organic, Precision

Identity: Three contrasting farming ideas — **shifting cultivation**, **certified organic**, and **site-specific precision** inputs.

System	Core	Where / tool
Jhum (shifting)	Slash-burn, 1–3 yr crop, then fallow	**NE** (jhum); **podu / penda** (AP/Odisha); **bewar / dahiya / mashan** (MP–CG); **kumari** (W Ghats); **waltre** (SE Rajasthan); **poonam** (Kerala); **kuruwa** (Jharkhand); **khil** (Himalayan belt)
Organic	No synthetic agrochemicals (standards)	Compost, rotation, Azolla, Rhizobium; **Sikkim** = first fully organic state (declared **2016**)
Precision	Site-specific inputs	GIS / GPS / sensors / drones — not “stop irrigation”

Jhum / shifting cultivation

Jhum is the north-east name for slash-and-burn shifting cultivation.

Farmers clear a forest patch, burn the residue, raise crops for about **one to three years**, then leave the plot fallow while opening a new patch.

Short fallows raise **erosion** and forest damage.

Regional names must be matched separately.

Podu / penda is the Andhra–Odisha label.

Bewar / dahiya / mashan is the Madhya Pradesh–Chhattisgarh label.

Kumari is the Western Ghats label.

Waltre is the south-east Rajasthan label.

Poonam is the Kerala label.

Kuruwa is the Jharkhand label.

Khil is the Himalayan-belt label.

Organic farming

Organic farming bans synthetic fertilisers and pesticides under a certification standard.

It relies on compost, crop rotation, green manure, **Azolla**, and **Rhizobium**-type biofertilisers.

Organic is not automatic for "any traditional practice."

Sikkim is India's first fully organic state, declared in **2016**.

NPOP is implemented with **APEDA** as secretariat.

The Ministry of Rural Development is **not** the NPOP operator.

Precision farming

Precision farming means site-specific input doses using **GIS, GPS, sensors and drones**.

It adjusts water, seed and fertiliser plot by plot.

It does **not** mean "stop irrigation" or abandon mechanisation.

N.10 MSP & Agricultural Marketing

- The **CACP** recommends MSP. The **Union Cabinet** decides. Do not swap CACP with RBI, NITI (नीति) Aayog (नीति आयोग), or the Finance Ministry.
- MSP covers **22 mandated crops**, plus derived cases such as toria and de-husked coconut.
- Procurement makes MSP real. Coverage is deepest for **wheat and paddy**.
- Pulses and oilseeds have weaker effective procurement.
- **PM-AASHA** supports price support for pulses and oilseeds through agencies such as NAFED.
- Sugarcane uses **FRP** (Fair and Remunerative Price), not the cereal MSP label.
- An **APMC** is a regulated mandi (मंडी).
- **e-NAM** is the electronic national mandi network.
- **PM-KISAN** gives direct income support to farmers.
- **PMFBY** is the crop-insurance scheme.
- **NFSM** pushes rice, wheat, pulses, millets and oilseeds.
- **NMSA** targets climate-smart farming and soil–water care.
- Budget **2021–22** placed an Agriculture Infrastructure and Development Cess (AIDC) on **29** products.
- Older paddy MSP figures such as **₹1,750/q** (2018) are year-specific. Keep the CACP-recommend / Cabinet-decide process, not the old rupee number.
- Fertiliser, power and irrigation support count as **indirect** subsidy.

PYQ — UPPCS Prelims 2024, Q43

Who recommends the Minimum Support Price (MSP) for agricultural crops?

A. RBI

B. NITI Aayog

C. CACP

D. Ministry of Finance

Show answer

Ans: C — CACP recommends. The Union Cabinet decides.

N.11 Green Revolution & Colour Revolutions

Green Revolution (हरित क्रांति) (mid-1960s): **HYV + irrigation + fertiliser** (+ pesticide, credit, machines).

- The Green Revolution heartland is **Punjab, Haryana and western UP**.
- It started as a **wheat** revolution, then expanded to rice. Some notes also tag maize, soybean and cane yield jumps.
- Classic HYVs include Mexican wheat **Lerma Rojo** and **Sonora 64** from **CIMMYT**, and rice **IR-8** from IRRI.
- Rockefeller Foundation support is a taught funding tag.
- Phasing is often taught as first stage about **1966–1981**, second **1981–1995**, and third from **1995** outward.
- Gains included a yield surge, surplus, and exit from PL-480 dependence.
- Wheat gained most in production and productivity; rice came next.
- Costs included regional inequality, a **groundwater** crash, nutrient mining, millet and pulse neglect, rice–wheat **monoculture**, and stubble burning.
- Social and environmental costs are real — do not treat “no cost” as true.
- Large parts of **rainfed / dry-zone** India are still **agrarian**: farming remains the main livelihood even where rainfall is low.
- Those same tracts can raise yields sharply if **irrigation** arrives. That is the **second Green Revolution** idea for drylands the first Green Revolution skipped.
- Second Green Revolution aims to extend seed–water–fertiliser to left-out areas and integrate crops with animal husbandry, social forestry (सामाजिक वानिकी) and fishing — not only more wheat–rice in already-benefited belts.
- **BGREI** (Bringing Green Revolution to Eastern India) sits under RKVY.
- It targets **rice-based** systems in the east and unused water, not a second Punjab in the desert.
- **Evergreen Revolution** (Swaminathan) means high productivity **without** ecological harm.
- Steps taught include biofertiliser/compost with chemicals, rainwater harvesting, agri-economic zones and contract farming.
- **Rainbow Revolution** entered the **National Agricultural Policy (28 July 2000)** as an integrated colour-revolution package.
- The National Zero Till Seed-cum-Fertilizer Drill from Pantnagar links Green Revolution residue management to zero tillage.

Person	Role
M.S. Swaminathan	GR in India; Evergreen idea
Norman Borlaug	HYV wheat; Nobel **Peace** not Agriculture

Person	Role
Verghese Kurien	White / milk
L.K. Jha	Economic Administration Reforms
Morarji Desai (आमिल)	Bank social control
Amartya Sen	Food **entitlements** — not Swaminathan
Hiralal Chaudhuri	Blue Revolution (fisheries) tag
Revolution	Sector
----------	--------
Green	Food grains (wheat → rice)
White	Milk (Kurien / Flood / NDDB)
Blue	Fisheries / aquaculture
Yellow	Oilseeds
Golden	Horticulture **and honey**
Silver	Eggs / poultry
Pink	Onion / meat / prawn (context)
Grey	Fertilisers
Black	Petroleum
Brown	Sometimes leather / cocoa / biofuel talk — weaker association
Rainbow	Integrated multi-sector
Evergreen	Sustainable productivity

Colour Revolutions — must-score set

The colour tags name sector pushes that sit beside the classic Green Revolution.

The **Golden Revolution** covers **horticulture and honey**.

The **Grey Revolution** covers **fertilisers**.

The **Yellow Revolution** covers **oilseeds**.

The **Black Revolution** covers **petroleum**.

Do **not** swap Golden with Yellow. Golden is horticulture and honey; Yellow is oilseeds.

Colour	Sector
Golden	Horticulture **and honey**
Grey	Fertilisers

Colour	Sector
Yellow	Oilseeds
Black	Petroleum

UP Focus (compact)

Identity: Uttar Pradesh is a **plains agricultural giant** — wheat–rice–sugarcane in the west and centre, rice in the east, potato and mango as specialty tags, and almost no plantation leadership.

Theme	Fact
Sugarcane	Quantity leader often; productivity < MH; Sugar Bowl tag
Wheat–rice	Western UP GR (Upper Gangetic / Zone 5)
Potato	Leading producer; CIP **Agra** (आगरा) not Aligarh
Mango	Dasheri / Langra; Amrapali hybrid
Dairy	Large bovine base; Jamunapari goat; top milk state
Mentha oil	UP is the classic near-total producer
Not	Jute / tea / rubber leader
SAU	**Pantnagar (first, 1960)**; Kanpur / Meerut

Uttar Pradesh often leads India in **sugarcane quantity**, but **Maharashtra** usually beats it on cane **productivity**.

Western Uttar Pradesh is the State's **Green Revolution** wheat–rice–sugarcane heartland (Upper Gangetic / Agro-climatic Zone 5).

Uttar Pradesh is India's leading **potato** producer.

The International Potato Centre (CIP) India office tag is **Agra**, not Aligarh.

Dasheri and **Langra** are classic UP mango identities.

Amrapali is a famous hybrid mango associated with Indian breeding notes.

Uttar Pradesh has a very large bovine base and is routinely among the top **milk** states.

The **Jamunapari** goat is an Etawah (इटावा)–Bundelkhand identity.

Mentha (mint) oil production is classically concentrated in Uttar Pradesh.

Uttar Pradesh is **not** India's jute, tea or rubber leader.

G.B. Pant University of Agriculture and Technology, Pantnagar (1960) is India's first agricultural university.

Kanpur and Meerut also host major State agricultural university campuses.